

What rule of inference is used in each of these arguments?

- Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials.
- It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous.
- Linda is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard.
- Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum.
- If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material.

- Simplification
- Disjunctive syllogism
- Modus ponens
- addition
- Hypothetical syllogism

5. Use rules of inference to show that the hypotheses "Randy works hard," "If Randy works hard, then he is a dull boy," and "If Randy is a dull boy, then he will not get the job" imply the conclusion "Randy will not get the job."

$P = \text{Randy works hard}$
 $Q = \text{Randy is a dull boy}$
 $R = \text{Randy will not get the job}$

P
 $P \rightarrow Q$ Modus ponens
 $\therefore Q$
 $Q \rightarrow R$ Modus ponens
 $\therefore R$

10. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

- "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool."
- "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."
- "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."
- "Every student has an Internet account." "Homer does not have an Internet account." "Maggie has an Internet account."
- "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat."
- "I am either dreaming or hallucinating." "I am not dreaming." "If I am hallucinating, I see elephants running down the road."

- We can conclude hockey was not played, and he is not sore, modus tollens shows us so, again shows not hockey.
- Sunny on Friday $\wedge$ (Sunny on Monday $\vee$ Partly sunny on Monday) Disjunctive syllogism
- Spiders are not insects, using modus tollens
- Homer is not a student, modus tollens.
- Tofu does not taste good. Modus ponens
- They see elephants running down the road. Disjunctive syllogism, modus ponens

16. For each of these arguments determine whether the argument is correct or incorrect and explain why.

- Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
- A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.
- Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
- All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps.

- Correct, $\forall x (enrolled(x) \rightarrow dorm(x))$, $\neg dorm(mia)$
 $\therefore \neg enrolled(mia)$
- incorrect, $convertible \rightarrow fun$, $\neg convertible \not\rightarrow \neg fun$
- incorrect, fallacy of affirming conclusion
- correct

19. Determine whether each of these arguments is valid. If an argument is correct, what rule of inference is being used? If it is not, what logical error occurs?

- If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$. Then $n > 1$.
- If n is a real number with $n > 3$, then $n^2 > 9$. Suppose that $n^2 \leq 9$. Then $n \leq 3$.
- If n is a real number with $n > 2$, then $n^2 > 4$. Suppose that $n \leq 2$. Then $n^2 \leq 4$.

- invalid, fallacy of affirming conclusion
- correct, modus tollens
- invalid, given $p \rightarrow q$, $\neg q \not\rightarrow \neg p$

23. Identify the error or errors in this argument that supposedly shows that if $\exists x P(x) \wedge \exists x Q(x)$ is true then $\exists x (P(x) \wedge Q(x))$ is true.

- $\exists x P(x) \vee \exists x Q(x)$ Premise
- $\exists x P(x)$ Simplification from (1)
- $P(c)$ Existential instantiation from (2)

4. $\exists xQ(x)$ Simplification from (1)
5. $Q(c)$ Existential instantiation from (4)
6. $P(c) \wedge Q(c)$ Conjunction from (3) and (5)
7. $\exists x(P(x) \wedge Q(x))$ Existential generalization

1. premise is written w/ $\vee$ instead of $\wedge$

4. simplification is illeac for a conjunction, not disjunction.

7. it is not a given that $P(x) \wedge Q(x)$ are true for c.

24. Identify the error or errors in this argument that supposedly shows that if $\forall x(P(x) \vee Q(x))$ is true then $\forall xP(x) \vee \forall xQ(x)$ is true.
 1. $\forall x(P(x) \vee Q(x))$ Premise
 2. $P(c) \vee Q(c)$ Universal instantiation from (1)
 3. $P(c)$ Simplification from (2)
 4. $\forall xP(x)$ Universal generalization from (3)
 5. $Q(c)$ Simplification from (2)
 6. $\forall xQ(x)$ Universal generalization from (5)
 7. $\forall x(P(x) \vee \forall xQ(x))$ Conjunction from (4) and (6)

7. simplification is for a conjunction
 the correct is as follows

1. 7

2. Use a direct proof to show that the sum of two even integers is even.

An integer is even if, for integer k,
 then exists an integer a for which $k=2a$ is true.

Let a, b be even integers

$$a=2c, b=2d$$

$$2(+2d)=2(c+d)$$

$$a+b=2(c+d)$$

therefore, by def., $a+b$ is even

5. Prove that if $m+n$ and $n+p$ are even integers, where m, n , and p are integers, then $m+p$ is even. What kind of proof did you use?

Let m,n,p be integers such that
 $m+n, n+p$ are even integers.

for $m+n$ to be even, m,n must have the same parity,
 i.e. m,n are both odd or n,n are both even. the same applies
 to $n+p$. for both possibilities the two, m,n,p, must be of the
 same parity. therefore, as m,p are of the same parity,
 $m+p$ is even.

proof by exhaustion.

10. Use a direct proof to show that the product of two rational numbers is rational.

let there be two rational numbers, $\frac{a}{b}, \frac{c}{d}$.
 for them to be rational, $b,d \neq 0$.

As $b,d \neq 0, bd \neq 0$

therefore $\frac{ac}{bd}$, the product of $\frac{a}{b}, \frac{c}{d}$, is rational.

17. Use a proof by contraposition to show that if $x+y \geq 2$, where x and y are real numbers, then $x \geq 1$ or $y \geq 1$.

$$P(x,y) := x+y \geq 2$$

$$q(x) := x \geq 1$$

$$s(y) := y \geq 1$$

$$\text{premise : } \sim (q(y) \vee s(y)) \rightarrow \sim P(x,y)$$

$$(\sim q(x) \wedge \sim s(y)) \rightarrow \sim P(x,y)$$

let a,b be real numbers such that $0 < a < 1$
 $0 < 1 + b < 1 = a+b < 2$

$$(\sim q(a) \wedge \sim s(b)) \rightarrow \sim P(a,b) \equiv T$$

$$\text{therefore, } P(x,y) \rightarrow (q(x) \vee s(y))$$

20. Prove that if n is an integer and $3n+2$ is even, then n is even using

- a) a proof by contraposition.
- b) a proof by contradiction.

$$P(n) := n \text{ in } \mathbb{Z}$$

$$q(n) := 3n+2 \text{ is even}$$

a)
 n is odd
 then exists k such that $n=2k+1$

$$3n+2 = 3(2k+1)+2 = 6k+3+2 = 6k+5 = 2(3k+2)+1$$

this fits the def. of an odd number,
 therefore n is odd $\rightarrow 3n+2$ is odd

b)
 assume $3n+2$ is even, n is odd

then exists k such that $n=2k+1$

$$3n+2 = 3(2k+1)+2 = 6k+3+2 = 6k+5 = 2(3k+2)+1$$

this fits the def. of an odd number,

therefore proving $3n+2$ is even $\rightarrow$ n is even by contradiction

28. Prove that if n is a positive integer, then n is even if and only if $7n+4$ is even.

n is odd
 then exists k such that $n=2k+1$

$$7(2k+1)+4 = 14k+7+4 = 14k+11 = 2(7k+5)+1$$

this fits the def. of an odd number,

therefore proving $7n+4$ is even $\rightarrow$ n is even by contraposition

n is even
 then exists k such that $n=2k$

$$7(2k)+4 = 14k+4 = 2(7k+2)$$

therefore n is even $\rightarrow 7n+4$ is even by direct proof

$$(n \text{ is even} \rightarrow 7n+4 \text{ is even}) \wedge (7n+4 \text{ is even} \rightarrow n \text{ is even})$$

$$\equiv n \text{ is even} \Leftrightarrow 7n+4 \text{ is even}$$

therefore, $n \text{ is even} \Leftrightarrow 7n+4 \text{ is even}$